

Lecture on
Waves & Oscillations:
Damped Harmonic Motion
By Md Asaduzzaman Asad
INS, United International University

Damped oscillations

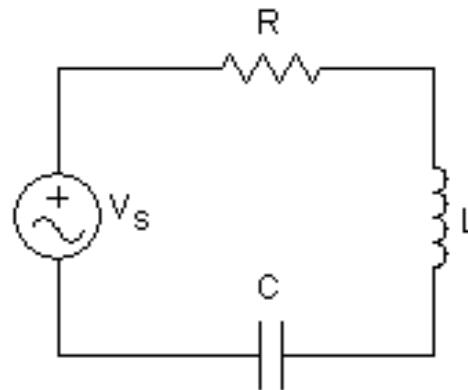
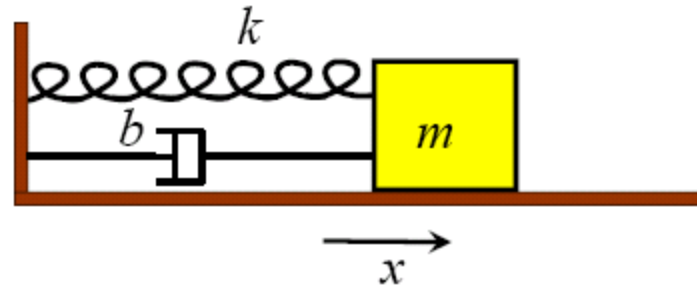
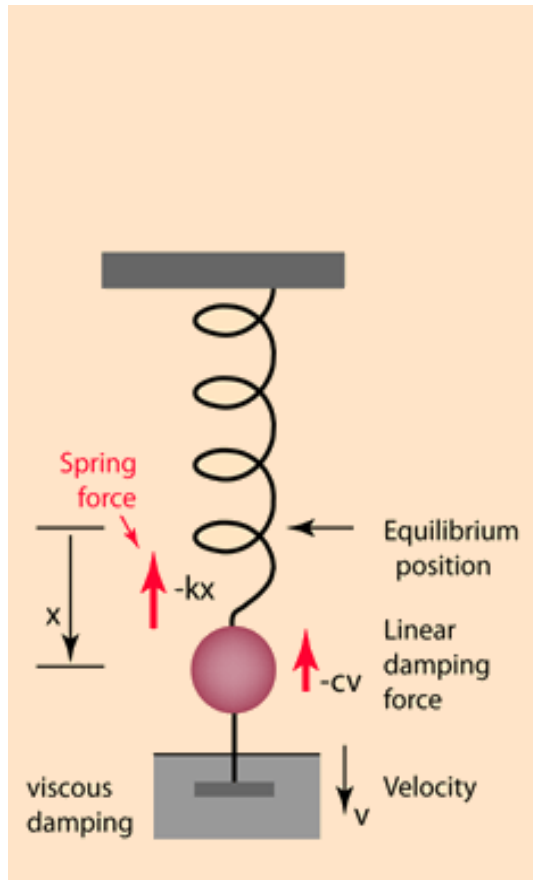
We have thus far neglected all dissipative mechanisms
... our oscillations can continue oscillating with the
same amplitude forever ...

Various physical damping mechanisms will contribute towards
the damping...

- friction between mass and table
- “air resistance”
- internal friction in spring
-

... model these by introducing a damping force which is
proportional to the velocity of the oscillator ...

Examples some damped oscillating systems





Damped Harmonic motion: When oscillating bodies do not move back and forth between precisely fixed limits because frictional force dissipates the energy and amplitude of oscillation decreases with time and finally die out. Such harmonic motion is called Damped Harmonic Motion.

Damped mass-spring system

In these systems the damping force

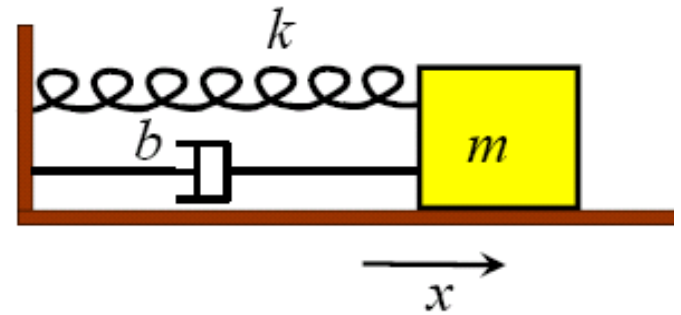
$$F' = -bv$$

For horizontal forces on the mass: $ma = -kx - bv$

$$\text{or } m \frac{d^2x}{dt^2} = -kx - b \frac{dx}{dt}$$

$$\text{or } \frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0 \quad \text{where } \begin{cases} \omega_0 = \sqrt{\frac{k}{m}} \\ \gamma = \frac{b}{m} \end{cases}$$

γ : "damping constant" unit: s^{-1} • "life time" = $\frac{1}{\gamma}$



Damped oscillator equation

$$\frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0$$

Let $x = Be^{pt}$

Then $\frac{dx}{dt} = Bpe^{pt}$ and $\frac{d^2x}{dt^2} = Bp^2e^{pt}$

Substituting into DE: $Bp^2e^{pt} + \gamma Bpe^{pt} + \omega_0^2 Be^{pt} = 0$

Thus $p^2 + \gamma p + \omega_0^2 = 0$

$$\therefore p = \frac{1}{2} \left\{ -\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2} \right\}$$

or $p = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Damped oscillator equation ...2

$$p = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

We can distinguish three cases:

- (i) $\omega_0^2 > \frac{\gamma^2}{4}$ **Oscillatory behaviour**
- (ii) $\omega_0^2 = \frac{\gamma^2}{4}$ **Critical damping**
- (iii) $\omega_0^2 < \frac{\gamma^2}{4}$ **Overdamping**

$$\text{Case (i):} \quad \omega_0^2 > \frac{\gamma^2}{4}$$

$$\therefore \sqrt{\gamma^2/4 - \omega_0^2} = \sqrt{-(\omega_0^2 - \gamma^2/4)}$$

$$\text{Put} \quad \omega_1^2 = \omega_0^2 - \gamma^2/4$$

$$\therefore p = -\frac{\gamma}{2} \pm \sqrt{-\omega_1^2} = -\frac{\gamma}{2} \pm j\omega_1$$

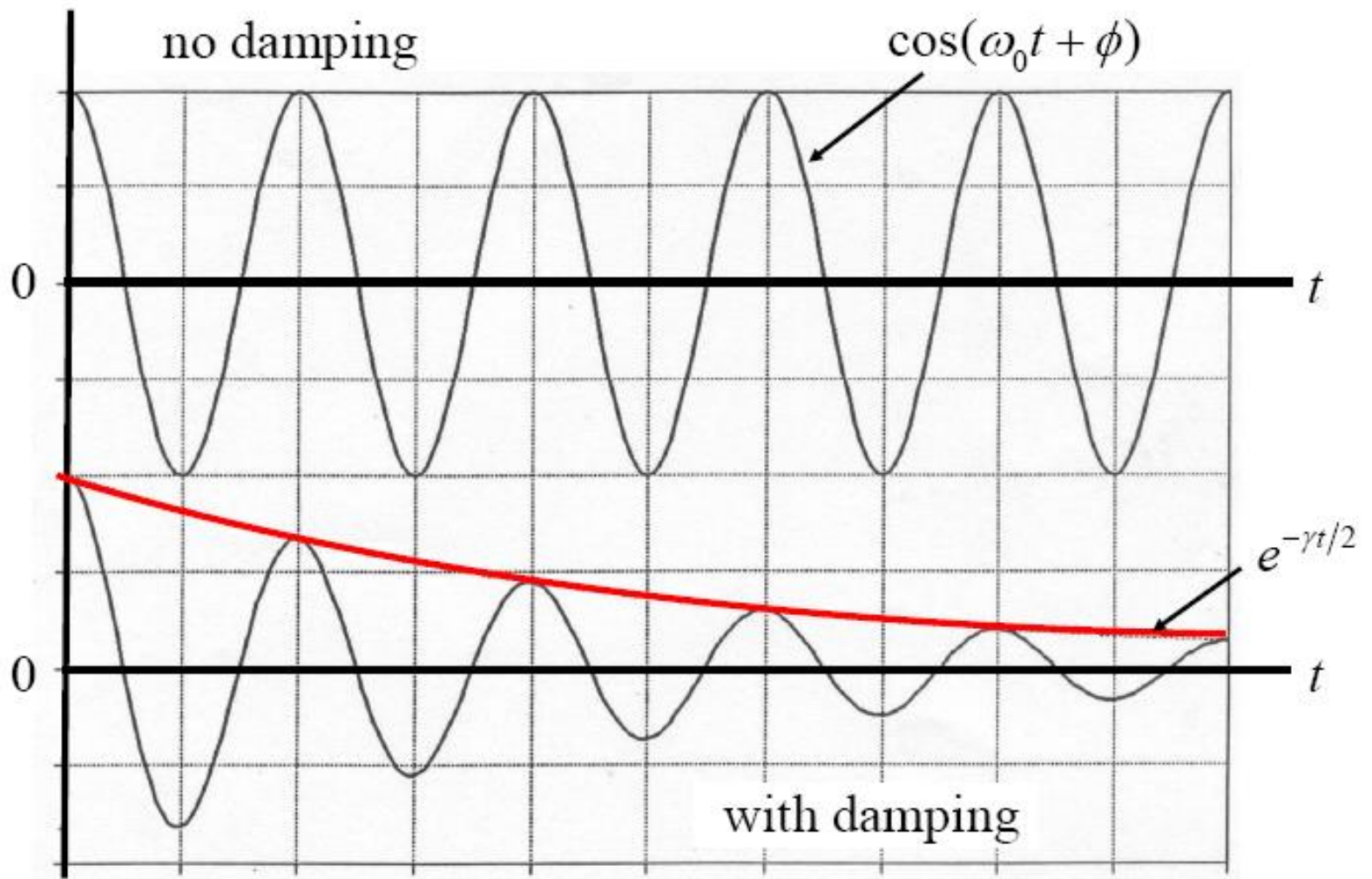
The solution will be:

$$x = B_1 e^{\left(-\frac{\gamma}{2} + j\omega_1\right)t} + B_2 e^{\left(-\frac{\gamma}{2} - j\omega_1\right)t} = e^{-\frac{\gamma}{2}t} \left\{ B_1 e^{j\omega_1 t} + B_2 e^{-j\omega_1 t} \right\}$$

$$\dots \text{ leading to } x(t) = A e^{-\frac{\gamma t}{2}} \cos(\omega_1 t + \phi)$$

This is an **oscillatory solution** $A \cos(\omega_1 t + \phi)$ multiplied by a damping factor $e^{-\gamma t/2}$.

As $\gamma \rightarrow 0$ we approach our undamped oscillator.



Case (ii): $\omega_0^2 = \frac{\gamma^2}{4}$

The two roots coincide: $p = -\frac{\gamma}{2}$

The solution will be $x(t) = (A + Bt)e^{-\frac{\gamma}{2}t}$

The condition $\omega_0^2 = \gamma^2/4$ is referred to as the “**critical damping**” condition.

If $\omega_0^2 < \gamma^2/4$ a system released from rest will oscillate.

As γ is increased the oscillations decay more rapidly, until at $\omega_0^2 = \gamma^2/4$ oscillation no longer occurs.

[... many practical applications ...]

$$\text{Case (iii): } \omega_0^2 < \frac{\gamma^2}{4}$$

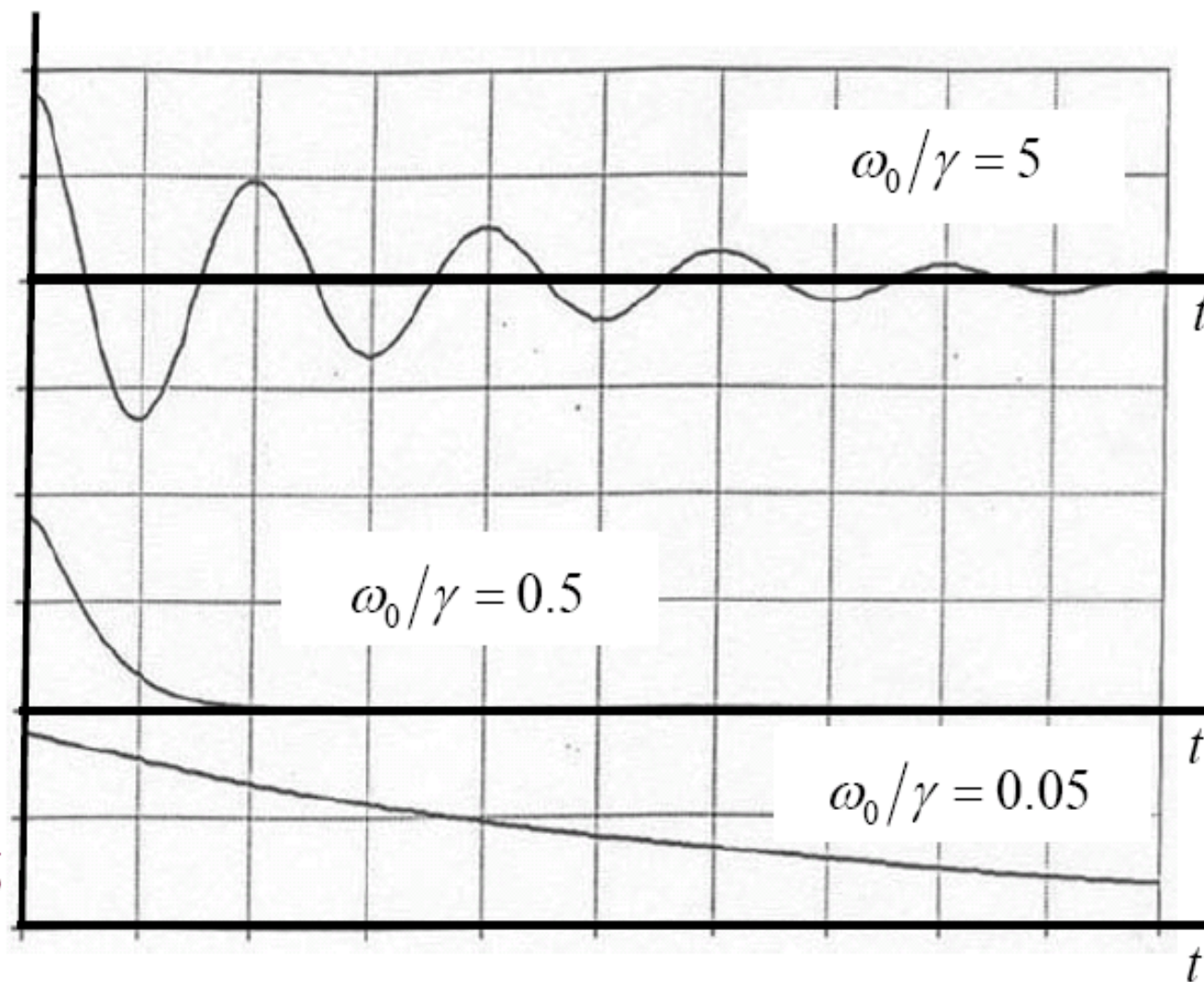
$$\begin{aligned} p &= -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2} \\ &= -\frac{\gamma}{2} \pm \lambda \quad \text{say} \end{aligned}$$

The solution will be $x(t) = B_1 e^{(-\frac{\gamma}{2} + \lambda)t} + B_2 e^{(-\frac{\gamma}{2} - \lambda)t}$

The condition $\omega_0^2 < \frac{\gamma^2}{4}$ is referred to as **overdamping**

... a slower approach to the rest position is observed.

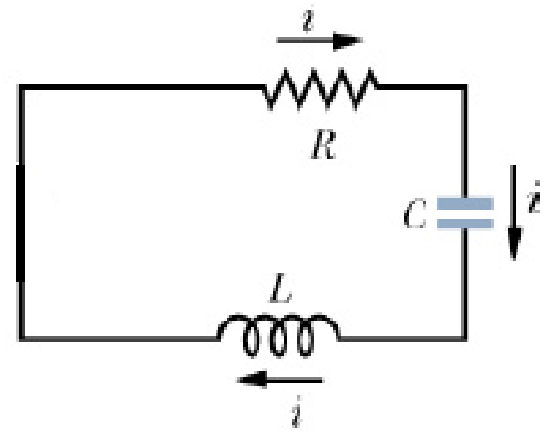
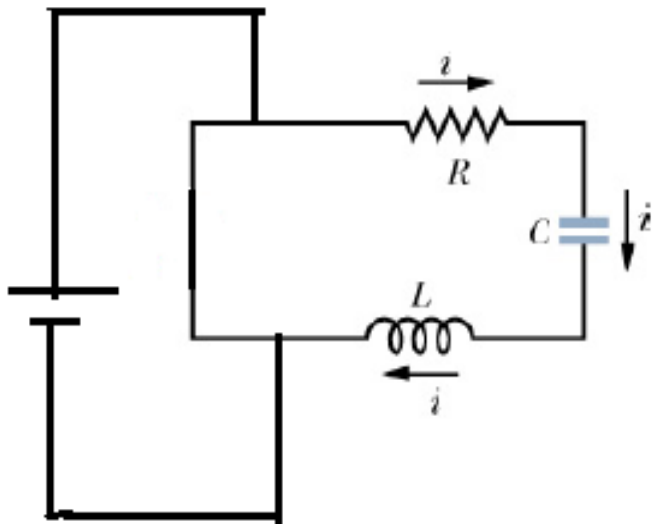
Oscillatory



**Critical
damping**

Overdamping

RLC circuit



- Voltage across resistor R $V_R = iR$
- Voltage across capacitor C $V_C = \frac{Q}{C}$
- Voltage across inductor L $V_L = L \frac{di}{dt}$
- According to Kirchhoff's voltage law

$$iR + \frac{Q}{C} + L \frac{di}{dt} = 0$$

Rewrite the equation

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = 0$$

Comparing with the equation

$$\frac{d^2 x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0$$

Where

$$\gamma = \frac{R}{L} \quad \omega_0 = \sqrt{\frac{1}{LC}}$$

Three distinguish cases are

i) $\frac{1}{LC} > \frac{R^2}{4L^2}$ Oscillatory behavior

ii) $\frac{1}{LC} = \frac{R^2}{4L^2}$ Critical damping

iii) $\frac{1}{LC} < \frac{R^2}{4L^2}$ Over damping

Case i) $\frac{1}{LC} > \frac{R^2}{4L^2}$

Solution of the differential equation

$$Q(t) = Ae^{-\frac{R}{2L}t} \cos(\omega_1 t + \phi)$$

Where $\omega_1 = \sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2}\right)}$

Frequency of oscillation

$$f = \frac{1}{2\pi} \sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2}\right)}$$

EX-15: A capacitor $1.0\mu\text{F}$, an inductor 0.2h and a resistance 800Ω are joined in series. Is the circuit oscillatory? Find the frequency of oscillation.

Ex-16: Find whether the discharge of capacitor through the following inductive circuit is oscillatory.
 $C = 0.1\mu\text{F}$, $L = 10\text{mh}$, $R = 200\Omega$
If Oscillatory, find the frequency of oscillation.

Damped Harmonic Motion



Thank You

The END